

02

Digital Electronics & VLSI
Codec Technologies India
Internship 2026

Low-Power 8-bit ALU in CMOS

Clock gating + operand isolation
vs free-running baseline,
gate-level power

AT A GLANCE

63.8 %
Total power reduction @ 29 %
active

390.8 / 141.3 uW
Baseline / low-power P_{total}

275 741 / 119 277
Net toggles (same stimulus)

2442 / 2446
Cell area (um²)

2.13 / 2.70 ns
Critical path (mini_sta)

2460 / 2450
Transistors (estimate)

4 x 590 pass
Gate-level checks

80.5 % / 8.3 %
Reduction @ 10 % / 100 % active

VERDICT

RTL PASS GATE-LEVEL PASS

POWER -63.8 %

DECLARED SUBSTITUTION

1. Objective and brief mapping

The brief asked for an 8-bit ALU (ADD, SUB, AND, OR, XOR, NOT, SHL, SHR, INC, DEC, CMP with Z, C, N, V flags) in two variants, a free-running baseline and a low-power version using a latch-free ICG, operand isolation and one-hot decode; both mapped to the project CMOS cell library with yosys, gate-level simulated with the same random stimulus at about 30 % activity, then compared with power_estimate.py and mini_sta.py. It also asked for a transistor-level view (schematics, stick diagram, transistor count table), an ngspice-compatible SPICE deck of a full adder and a Microwind/Magic how-to, with the percentage power reduction stated from actual numbers.

Everything numeric in this report comes from runs in the build container: Icarus Verilog 14 for RTL and gate-level simulation, yosys 0.68 for mapping, and the repository's Python estimators. SPICE and layout tools were not available; their input files are supplied and declared in section 6.

2. Architecture of the two variants

Both variants share the interface (clk, rst, en, a, b, op -> result, flags, valid) and the two-cycle latency, so the testbench and the reference model are identical. In the baseline the operand bus is registered every cycle and all four functional units switch with it, even when en is low. In the low-power variant the operand registers only receive a clock edge when en is high, the result registers only when a result is due (en_q), and each unit sees zeros on its inputs unless its opcode group is selected. The one-hot decode replaces the priority case-mux by an AND-OR merge, which also removes a glitch-prone select chain.

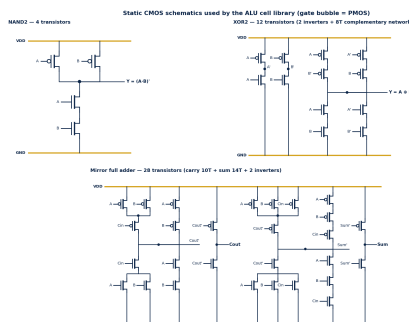
baseline: en,a,b,op -> [34 DFF on clk] -> ADD | SUB | AND/OR/XOR/NOT | SHL/SHR all evaluate ->

low-power: en -----> ICG_in (negedge FF + AND) -> gclk_in -> [a_q b_q op_q : 20 DFF]
en_q ----> ICG_out (negedge FF + AND) -> gclk_out -> [result flags : 12 DFF]
op_q -> one-hot s_add..s_cmp ; g_add = ADD|INC, g_sub = SUB|DEC|CMP, g_log = AND|OR|XOR
adder(a_q&g_add, b_q&s_add | s_inc) subtractor(a_q&g_sub, b_q&(s_sub|s_cmp) | s_dec)
logic(a_q&g_log, b_q&g_log) shifter(a_q&g_sh) -> AND-OR merge ->

Technique	Where	Cost	Effect measured
Clock gating (latch-free ICG)	operand bank (20 DFF) and result bank (12 DFF)	2 negedge DFF + 2 AND	clock-pin toggles 4013 x 34 -> 4013 x 4 + 1184 x 32
Operand isolation	AND gates on adder, subtractor, logic and shifter inputs	~40 AND2, +0.57 ns path	8.3 % saving even at 100 % activity
One-hot decode + AND-OR merge	opcode decode and result selection	neutral (fewer MUX2, more NAND/NOR)	no priority chain glitching

3. Implementation and flow

- icg_cell.v: en is sampled on the falling edge of clk, so en_q is stable during the high phase and gclk = clk & en_q either reproduces the full pulse or stays low; test_en forces the clock on for scan. yosys maps it to a DFF with an inverted clock plus an AND cell.
- Mapping script per variant (synth/map_*.ys): synth -flatten; delete t:\$scopeinfo; dfflibmap -liberty; abc -liberty; opt_clean; rename -enumerate; write_json; write_verilog -noattr; stat -liberty. Enable/reset flops become plain DFF plus logic because the library only has a D flip-flop.
- Gate-level simulation uses the written netlist with tools/cmos_cells.v; the VCD scope tb_alu.dut is passed to power_estimate.py, which matched 435/435 (baseline) and 428/428 (low-power) netlist nets to VCD signals.
- The gate-level baseline VCD is 2.35 MB, so both gate-level VCDs are deleted after the power run; the RTL VCDs (0.7 MB) are kept for the waveform PNGs.
- cmos/draw_cmos.py draws the NAND2 (4T), the 12T XOR and the 28T mirror full adder as transistor schematics, a single-row stick diagram of the adder, and derives the transistor count of each netlist from the yosys cell counts.



cmos/schematics.png: static CMOS NAND2 (4T), XOR2 (12T) and the 28T mirror full adder whose SPICE deck is spice/full_adder_28t.sp.

4. Verification

tb_alu.v drives seeded random a, b, op on every cycle (the bus keeps toggling while idle, as a shared operand bus would) and asserts en with 30 % probability; 5 % of opcodes are illegal (11-15) to check the default branch. A behavioural reference model predicts result and {Z, C, N, V}; a two-deep expectation pipeline compares every valid output, and the valid signal itself is compared cycle by cycle.

KEY TAKEAWAY Gating two register banks with a latch-free ICG and isolating idle units cut measured switching power by 63.8 % at 29 % activity (80.5 % at 10 %) for 0.2 % more area and a 27 % longer critical path.

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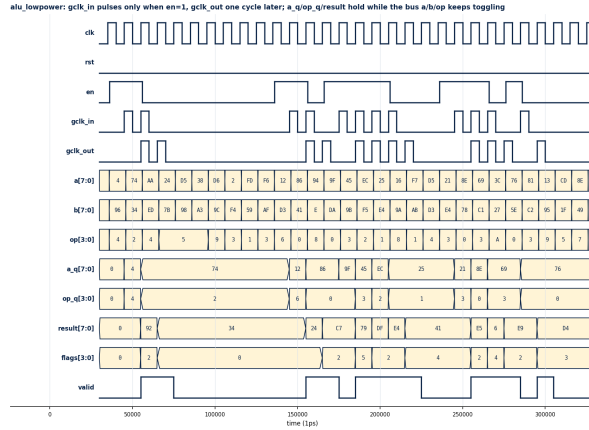
GATE-LEVEL PASS

POWER -63.8 %

DECLARED SUBSTITUTION

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Four runs, all PASS: RTL baseline, RTL low-power, gate-level baseline, gate-level low-power, each 2000 cycles, 590 active cycles, 590/590 matches, op histogram 38-59 per opcode.

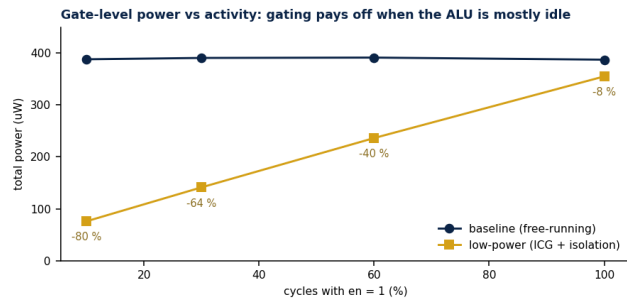


Low-power variant, RTL VCD: gclk_in pulses only in cycles where en was high, gclk_out one cycle later; a_q/op_q/result/flags hold their value while a/b/op keep changing.

5. Results: area, timing, power

Power by driving cell type shows where the saving comes from: DFF-driven nets fall from 63.7 to 21.7 uW, NOR from 96.5 to 18.9 uW and the primary-input/clock wires from 61.7 to 15.1 uW; the AND-driven nets stay at about 41-44 uW because the isolation gates themselves switch in active cycles. The delay penalty is the isolation AND on the operand path plus the ICG in the clock path (23 logic levels vs 20).

Metric	baseline	low-power	change
Cells (yosys stat)	412	405	-1.7 %
Cell area um ² (stat -liberty)	2442	2446	+0.2 %
Transistors (NAND/NOR 4, NOT 2, AND/OR 6, XOR/XNOR/MUX2 12, DFF 20)	2460	2450	-0.4 %
DFF count	34	36	+2
Critical path / Fmax (mini_sta)	2.133 ns / 468.8 MHz	2.702 ns / 370.2 MHz	+26.7 % delay
Net toggles in 20.06 us	275 741	119 277	-56.7 %
P _{dynamic} (1.8 V, 100 MHz)	390.45 uW	140.93 uW	-63.9 %
P _{leakage}	0.354 uW	0.346 uW	-2.3 %
P _{total}	390.80 uW	141.28 uW	-63.8 %



Same netlists, same seed, activity swept 10-100 %. The baseline is flat because it switches regardless of en; the low-power design approaches it only when every cycle is active.

5b. Transistor-level view and power breakdown

Transistor counts are derived from the yosys cell counts with the static-CMOS figures listed in the brief (NAND/NOR 4, NOT 2, AND/OR 6, XOR/XNOR/MUX2 12, DFF 20). The two designs are within 0.4 % of each other; the low-power version trades XOR/NOR cells for NAND/OR/MUX2 in the AND-OR merge and adds two DFFs for the ICG enables.

Cell	T/cell	baseline cells	baseline T	low-power cells	low-power T
NOT	2	16	32	20	40
NAND	4	121	484	140	560
NOR	4	148	592	122	488
AND	6	57	342	43	258
OR	6	17	102	24	144
XOR	12	6	72	2	24
XNOR	12	11	132	10	120
MUX2	12	2	24	8	96
DFF	20	34	680	36	720
total		412	2460	405	2450

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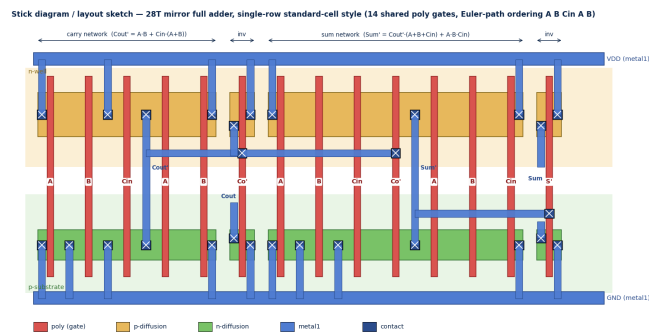
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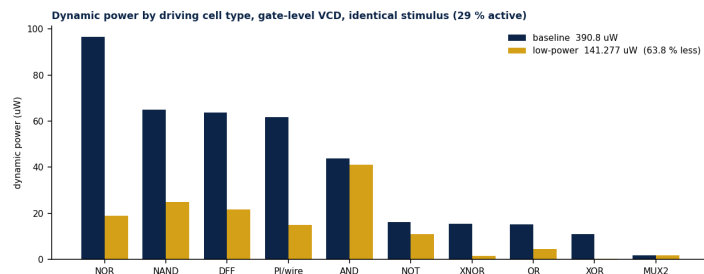
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Stick diagram of the 28T mirror adder as one standard-cell row: p-diffusion under the VDD rail, n-diffusion above GND, 14 shared poly gates in Euler-path order, metal1 for rails, Cout/Sum internal nodes and outputs.

Dynamic power by the type of cell driving each net (uW, 29 % activity). The DFF, NOR and primary-input rows carry the clock-gating saving; the AND row barely moves because the isolation gates switch whenever an operation runs.

Driving cell type	baseline uW	low-power uW
NOR	96.52	18.883
NAND	64.928	24.902
DFF	63.744	21.735
PI/wire	61.749	15.095
AND	43.767	41.002
NOT	16.19	11.046
XNOR	15.39	1.675
OR	15.183	4.571
XOR	11.083	0.302
MUX2	1.893	1.72



docs/power_compare.png: the same breakdown as a bar chart (power_estimate.py, identical stimulus).

6. Declared substitutions and target-tool artefacts

- ngspice / HSPICE / Spectre: not installed. spice/full_adder_28t.sp contains the 28-transistor mirror adder (14 PMOS, 14 NMOS), generic 180 nm BSIM3v3 level-49 parameters, 1.8 V supply, three-frequency input pulses, .tran and .measure statements for tpd and average power. It was not executed; no SPICE number appears in this report.
- Microwind / Magic / Virtuoso: not installed. cmos/MICROWIND_MAGIC_HOWTO.md lists the layout steps (rule file, diffusion widths, poly order from the stick diagram, extraction and LVS transistor-count check). The stick diagram PNG is a matplotlib drawing, not an extracted layout.
- The cell library is the educational tools/cmos_cells.lib (180 nm-class representative values); power_estimate.py is an activity-based CV²f estimate without glitch or short-circuit power and mini_sta.py uses a linear delay model. The figures are valid for relative comparison of the two variants under identical stimulus, which is how they are used.

7. Learnings and next steps

The clock network dominated idle power: 34 clock pins toggling 4013 times cost more than the whole datapath, which is why gating just two register banks gave most of the 63.8 %. Operand isolation is cheap in area but not in delay, so on a timing-critical path I would isolate only the wide adder/subtractor and leave the shifter free. Next steps: add a third gated bank for the flag register alone (flags are needed less often than results), try a latch-based ICG to remove the extra negedge flop, and run the same testbench through OpenROAD with a real PDK to replace the estimators with signoff-grade numbers.

A second observation from the duty sweep: the baseline's power is flat within 1 % from 10 % to 100 % activity (387-391 uW), which confirms that in a free-running design the data-dependent part of the switching is small compared with the clock tree and the register loading; any power figure quoted for such a block without stating its activity is therefore almost meaningless, and I have stated the activity next to every number in this report.

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